HEIGHTS AND TORSION ON ELLIPTIC CURVES, WITH SOLUTIONS

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Kronecker's theorem says that the elements of $\overline{\mathbb{Q}}^\times$ of height 1 are the roots of unity, the torsion of $\mathbb{G}_\mathrm{m}$. This worksheet proves its analogue for an elliptic curve, for the height h_x of the x-coordinate, assuming the duplication bound of [Sil09, Corollary VIII.6.4(b)]. The torsion points have bounded height, not height 0, and that is enough for finiteness. First come the valuations of a sum of several terms, and the shape of the coordinates of a rational point, over $\mathbb{Q}$ and over a number field: the first step of Silverman's proof of the Mordell–Weil theorem over $\mathbb{Q}$. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A number field k, elements $a, b \in k$ with $\Delta := -16(4a^3 + 27b^2) \neq 0$, and the elliptic curve $E_{a,b}\colon y^2 = x^3 + ax + b$ over k.
- The height H on $\mathbb{P}^N(\overline{\mathbb{Q}})$, and $h := \log H$. For $t \in \mathbb{Q}$, with $t = u/w$ in lowest terms, $H([t:1]) = \max(|u|, |w|)$. See [Sil09, §VIII.5].
- The x-coordinate $x\colon E_{a,b} \to \mathbb{P}^1$, with $x(O) = [1:0]$, and the height $h_x(P) := h(x(P))$ of a point $P \in E_{a,b}(\overline{\mathbb{Q}})$. So $h_x(O) = 0$.

How to use this. Allow about 10 minutes for Problem 1.2, 15 for Problem 2.2 and 20 for Problem 3.4.

1. Valuations of sums

The proofs below compare the valuations of the terms of a sum. This section makes the ultrametric inequality precise: when it is an equality, and what happens when it is not.

Definition 1.1 (Discrete valuations)**.** A discrete valuation on a field F is a surjective map $v\colon F \to \mathbb{Z} \cup \{\infty\}$ with $v(\alpha) = \infty$ only for $\alpha = 0$, $v(\alpha\beta) = v(\alpha) + v(\beta)$ and $v(\alpha + \beta) \geqslant \min(v(\alpha), v(\beta))$. Its valuation ring is $\mathcal{O}_v := \{\alpha \in F : v(\alpha) \geqslant 0\}$, a uniformizer is an element $\pi \in F$ with $v(\pi) = 1$, and the residue field is $\kappa := \mathcal{O}_v/\pi\mathcal{O}_v$, with reduction $\alpha \mapsto \bar{\alpha}$. Examples are v_p on $\mathbb{Q}$, and $v_\mathfrak{p}$ on a number field k, for a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$.

Problem 1.2 (The unique minimizer criterion)**.**
Let v be a discrete valuation on a field F, with uniformizer π and residue field κ. Let $\alpha_1, \ldots, \alpha_n \in F$, not all 0, $m := \min_i v(\alpha_i)$, and $I := \{i : v(\alpha_i) = m\}$.

(1) Show that $v(\alpha_1 + \cdots + \alpha_n) \geqslant m$, with equality if I has one element.

Solution. **The other terms add up to an element of valuation $> m$.** The inequality follows from the definition, by induction on n. For the equality, first let $n = 2$ and $v(\alpha_1) < v(\alpha_2)$. If $v(\alpha_1 + \alpha_2) > v(\alpha_1)$, then

$$v(\alpha_1) = v\big((\alpha_1 + \alpha_2) - \alpha_2\big) \geqslant \min\big(v(\alpha_1 + \alpha_2), v(\alpha_2)\big) > v(\alpha_1),$$

a contradiction, since $v(-\alpha_2) = v(\alpha_2)$. In general, let $I = \{1\}$. Then $v(\alpha_2 + \cdots + \alpha_n) \geqslant \min_{i \geqslant 2} v(\alpha_i) > m$, and the case $n = 2$ applies to α_1 and $\alpha_2 + \cdots + \alpha_n$.

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(2) Show that $v(\alpha_1+\cdots+\alpha_n)=m$ if and only if $\sum_{i\in I}\overline{\alpha_i/\pi^m}\neq 0$ in κ. Give an example with $\#I=2$ where the inequality is strict.

SOLUTION. **Divide by π^m and reduce: the terms outside I vanish in κ.** Let $\beta_i := \alpha_i/\pi^m$, so that $v(\beta_i)\geqslant 0$, with equality exactly for $i\in I$. Then $v(\alpha_1+\cdots+\alpha_n)=m+v(\beta_1+\cdots+\beta_n)$, and $\beta_1+\cdots+\beta_n\in\mathcal{O}_v$ has valuation 0 if and only if its reduction is nonzero. For $i\notin I$, $v(\beta_i)\geqslant 1$, so $\bar{\beta}_i=0$, and the reduction is $\sum_{i\in I}\bar{\beta}_i$. For an example, take $v=v_p$ and $1+(p-1)=p$: both terms have valuation 0, and their residues 1 and -1 cancel.

Remark. In particular, equality holds when I has one element, since then the sum of residues is a single nonzero residue. This is (1) again. The criterion is the non-archimedean form of the dominant term: for large $|x|$, a polynomial behaves like its leading term, up to a constant. Here the dominant term wins exactly.

2. The shape of a rational point

This section is the first step of the proof of [Sil09, Lemma VIII.4.2], which proves that h_x satisfies the hypotheses of the descent theorem over $\mathbb{Q}$. It turns the rational points of bounded height into a finite search over integers. The second part is the same argument over a number field, where ideals take the place of the integer d. The integrality of a,b is harmless: if u^4a, u^6b are integral, then $(x,y)\mapsto(u^2x,u^3y)$ is an isomorphism onto E_{u^4a,u^6b}, and it changes h_x by at most $2\log|u|$ for $u\in\mathbb{Z}$.

Definition 2.1 (Heights over a number field [Sil09, §VIII.5])**.** For $t\in k$,

$$H([t:1])^{[k:\mathbb{Q}]}=\prod_{v\in\mathcal{P}_k}\max(1,|t|_v)^{n_v},$$

with $n_v := [k_v:\mathbb{Q}_p]$ or $[k_v:\mathbb{R}]$. For the place of a nonzero prime ideal $\mathfrak{p}$ of $\mathcal{O}_k$, $|t|_v^{n_v}=N(\mathfrak{p})^{-v_\mathfrak{p}(t)}$, where $N(\mathfrak{p})$ is the order of $\mathcal{O}_k/\mathfrak{p}$. The denominator ideal of t is $\mathfrak{d}(t) := \prod_\mathfrak{p}\mathfrak{p}^{\max(0,-v_\mathfrak{p}(t))}$, an ideal of $\mathcal{O}_k$.

Problem 2.2 (The shape of a rational point)**.**

(1) Let $a,b\in\mathbb{Z}$, and $P\in E_{a,b}(\mathbb{Q})\smallsetminus\{O\}$. Show that $x(P)=n/d^2$ and $y(P)=m/d^3$ with $n,m,d\in\mathbb{Z}$, $d\geqslant 1$ and $\gcd(n,d)=\gcd(m,d)=1$, and that $H(x(P))=\max(|n|,d^2)$.

SOLUTION. **At a prime p with $v_p(x)<0$, $3v_p(x)=2v_p(y)$.** Write $x=x(P)$ and $y=y(P)$. Let p be a prime with $v_p(x)<0$. Since $a,b\in\mathbb{Z}$, $v_p(ax)\geqslant v_p(x)>3v_p(x)$ and $v_p(b)\geqslant 0>3v_p(x)$, so x^3 is the only term of smallest valuation in x^3+ax+b, and $2v_p(y)=v_p(x^3+ax+b)=3v_p(x)$ by Problem 1.2(1). Then $v_p(x)=-2r$ and $v_p(y)=-3r$ for some integer $r\geqslant 1$. If $v_p(x)\geqslant 0$, then $v_p(y^2)=v_p(x^3+ax+b)\geqslant 0$, so $v_p(y)\geqslant 0$. Let $d := \prod_p p^r$, the product over the primes with $v_p(x)<0$. Then $n := d^2x$ and $m := d^3y$ are integers prime to d. As $\gcd(n,d^2)=1$, the fraction n/d^2 is in lowest terms, and $H(x(P))=\max(|n|,d^2)$.

Remark. So the rational points of $E_{a,b}$ with $H(x)\leqslant B$ have $|n|\leqslant B$ and $d\leqslant\sqrt{B}$, and $m^2=n^3+and^4+bd^6$: a finite search. This is how the descent bound of [Sil09, Theorem VIII.3.1] becomes a list of generators of $E_{a,b}(\mathbb{Q})$.

(2) Let $a,b\in\mathcal{O}_k$, and $P\in E_{a,b}(k)\smallsetminus\{O\}$. Show that there is an ideal $\mathfrak{d}$ of $\mathcal{O}_k$ with $\mathfrak{d}(x(P))=\mathfrak{d}^2$ and $\mathfrak{d}(y(P))=\mathfrak{d}^3$, and that

$$H(x(P))^{[k:\mathbb{Q}]}=N(\mathfrak{d})^2\prod_{v|\infty}\max(1,|x(P)|_v)^{n_v}.$$

Check that for $k = \mathbb{Q}$ this is (1).

SOLUTION. **The argument of (1) works at each prime $\mathfrak{p}$.** Write $x = x(P)$ and $y = y(P)$. At a prime $\mathfrak{p}$ with $v_\mathfrak{p}(x) < 0$, the term x^3 is again the only one of smallest valuation, as $v_\mathfrak{p}(a), v_\mathfrak{p}(b) \geqslant 0$. So $v_\mathfrak{p}(x) = -2r_\mathfrak{p}$ and $v_\mathfrak{p}(y) = -3r_\mathfrak{p}$ with $r_\mathfrak{p} \geqslant 1$, and $v_\mathfrak{p}(y) \geqslant 0$ when $v_\mathfrak{p}(x) \geqslant 0$. Let $\mathfrak{d} := \prod_\mathfrak{p} \mathfrak{p}^{r_\mathfrak{p}}$, over the primes with $v_\mathfrak{p}(x) < 0$. Then $\mathfrak{d}(x) = \mathfrak{d}^2$ and $\mathfrak{d}(y) = \mathfrak{d}^3$. In Definition 2.1, the factor of a prime $\mathfrak{p}$ is $\max(1, N(\mathfrak{p})^{-v_\mathfrak{p}(x)}) = N(\mathfrak{p})^{\max(0,-v_\mathfrak{p}(x))}$, so the product of these factors is $N(\mathfrak{d}(x)) = N(\mathfrak{d})^2$. For $k = \mathbb{Q}$, $\mathfrak{d} = d\mathbb{Z}$ with $d \geqslant 1$, $N(\mathfrak{d}) = d$, and the one archimedean factor is $\max(1, |n|/d^2)$, so $H(x) = d^2 \max(1, |n|/d^2) = \max(|n|, d^2)$.

Remark. If $\mathfrak{d} = d\mathcal{O}_k$ is principal, then $x = n/d^2$ and $y = m/d^3$ with $n := d^2x$ and $m := d^3y$ in $\mathcal{O}_k$, as over $\mathbb{Q}$. In general $\mathfrak{d}$ need not be principal, and the class group is the obstruction to writing x this way. The height still bounds $N(\mathfrak{d})$, which leaves finitely many ideals $\mathfrak{d}$.

3. Kronecker's theorem for elliptic curves

Kronecker's theorem [Sil09, Exercise 8.6] says that $\alpha \in \overline{\mathbb{Q}}^\times$ has $H(\alpha) = 1$ if and only if α is a root of unity. This section proves the analogue for $E_{a,b}$ and h_x, from the results below.

Proposition 3.1 (The duplication formula [Sil09, Algorithm III.2.3(d)])**.** *Let $P \in E_{a,b}(\overline{\mathbb{Q}})$ with $2P \neq O$, and $x := x(P)$. Then*

$$x(2P) = \frac{x^4 - 2ax^2 - 8bx + a^2}{4(x^3 + ax + b)}.$$

Proposition 3.2 (The duplication bound [Sil09, Corollary VIII.6.4(b)])**.** *There is a constant $C \geqslant 0$, depending on a and b, with $|h_x(2P) - 4h_x(P)| \leqslant C$ for all $P \in E_{a,b}(\overline{\mathbb{Q}})$.*

Theorem 3.3 (Northcott [Sil09, Theorem VIII.5.11])**.** *For all constants B and D, the set of $R \in \mathbb{P}^N(\overline{\mathbb{Q}})$ with $H(R) \leqslant B$ and $[\mathbb{Q}(R) : \mathbb{Q}] \leqslant D$ is finite. Here $\mathbb{Q}(R)$ is the field of definition of R.*

Problem 3.4 (Kronecker for elliptic curves)**.**
Let C be the constant of Proposition 3.2. Only its lower bound, $h_x(2P) \geqslant 4h_x(P) - C$, is used.

(1) Show that every torsion point $T \in E_{a,b}(\overline{\mathbb{Q}})$ has $h_x(T) \leqslant C/3$.

SOLUTION. **The largest height M on the group $\langle T \rangle$ has $4M - C \leqslant M$.** Let $M := \max\{h_x(P) : P \in \langle T \rangle\}$, attained at nT for some $n \in \mathbb{Z}$. Then $2nT \in \langle T \rangle$, so

$$4M - C = 4h_x(nT) - C \leqslant h_x(2nT) \leqslant M,$$

and $M \leqslant C/3$. In particular, $h_x(T) \leqslant M \leqslant C/3$.

Remark. The argument uses only two properties of $\langle T \rangle$: it is finite, and it contains the double of each of its elements. The orbit $\{2^jT : j \geqslant 0\}$ of T under doubling has both, and would do as well.

(2) Show that for every $D \geqslant 1$ there are only finitely many torsion points $T \in E_{a,b}(\overline{\mathbb{Q}})$ with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Deduce that the torsion subgroup of $E_{a,b}(K)$ is finite for every number field K containing k.

SOLUTION. **Their x-coordinates have bounded height and bounded degree.** Let $T \neq O$ be a torsion point with $[\mathbb{Q}(T) : \mathbb{Q}] \leqslant D$. Then $x(T) \in \mathbb{Q}(T)$, so $[\mathbb{Q}(x(T)) : \mathbb{Q}] \leqslant D$, and $H(x(T)) \leqslant e^{C/3}$ by (1). By Theorem 3.3, only finitely many points of $\mathbb{P}^1(\overline{\mathbb{Q}})$ satisfy both conditions, and each is $x(T)$ for at most two points T. For a number field $K \supseteq k$, every $T \in E_{a,b}(K)$ has $\mathbb{Q}(T) \subseteq K$, so $D = [K : \mathbb{Q}]$ works.

Remark. For $\mathbb{G}_{\mathrm{m}}$ the same argument, with $h(\alpha^2) = 2h(\alpha)$ in place of Proposition 3.2, gives $h(\zeta) = 0$ for every root of unity ζ, and the finiteness of the roots of unity of bounded degree. On $E_{a,b}$, the error C leaves room for torsion points of positive height.

(3) Show that the naive analogue of Kronecker's theorem fails in both directions: a torsion point need not have $h_x = 0$, and a rational point with $h_x = 0$ need not be a torsion point.

(a) On $y^2 = x^3 + 1$, show that $T = (2, 3)$ is a torsion point of order 6, with $h_x(T) = \log 2$.

SOLUTION. **Here $x(2T) = 0$ and $x(4T) = 0$, so $4T = -2T$.** By Proposition 3.1 with $a = 0$ and $b = 1$, $x(2T) = (16 - 16)/(4 \cdot 9) = 0$, so $2T = (0, \pm 1)$, and $x(4T) = 0/4 = 0$. So $4T = \pm 2T$. As $y(T) \neq 0$, $2T \neq O$, so $4T \neq 2T$, and $4T = -2T$, that is, $6T = O$. The order is not 1, 2 or 3: $T \neq O$, $2T \neq O$, and $3T = O$ would mean $x(2T) = x(T) = 2$. So T has order 6, and $h_x(T) = \log H(2) = \log 2$.

(b) On $y^2 = x^3 + x + 1$, the rational point $P = (0, 1)$ has $h_x(P) = 0$. For this curve over $\mathbb{Q}$, one can take $C = \log 619$ in the lower bound $h_x(2P) \geqslant 4h_x(P) - C$. Show that P has infinite order.

SOLUTION. **Here $x(4P) = -287/1296$, and $1296 > 619^{1/3}$.** By Proposition 3.1 with $a = b = 1$, $x(2P) = 1/4$, and

$$x(4P) = \frac{(1/4)^4 - 2(1/4)^2 - 8(1/4) + 1}{4\big((1/4)^3 + 1/4 + 1\big)} = \frac{-287/256}{81/16} = -\frac{287}{1296},$$

in lowest terms, as $287 = 7 \cdot 41$. If P were a torsion point, so would be $4P$, and (1), which uses only the lower bound, would give $H(x(4P)) \leqslant 619^{1/3} < 9$. But $H(x(4P)) = 1296$. So P has infinite order.

Remark. The constant $\log 619$ comes from the explicit identities of [Sil09, Sublemma VIII.4.3], as in the proof of [Sil09, Lemma VIII.4.2]. The Lutz–Nagell theorem [Sil09, Corollary VIII.7.2] gives another proof: $x(2P) = 1/4$ is not an integer, so $2P$ is not a torsion point.

Remark. The canonical height $\hat{h}(P) := \lim_{n\to\infty} 4^{-n} h_x(2^n P)$ gives the exact analogue of Kronecker's theorem: $\hat{h}(mP) = m^2 \hat{h}(P)$, and $\hat{h}(P) = 0$ if and only if P is a torsion point [Sil09, Theorem VIII.9.3(b), (d)], where $\hat{h}$ is half of ours. Writing $\hat{h} - h_x$ as a telescoping series with terms $4^{-n-1}\big(h_x(2^{n+1}P) - 4h_x(2^n P)\big)$ gives $|\hat{h} - h_x| \leqslant C/3$. So (1) says that $h_x(T)$ is within $C/3$ of $\hat{h}(T) = 0$.

REFERENCES

[Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 ↑